

Nebula Mind

Nebula Mind is an advanced AI-powered "AI Powered Learning" application designed to revolutionize how you learn and organize information. Built with Next.js 15 and powered by cutting-edge AI models (Gemini, OpenAI, Ollama), it transforms your static documents into interactive knowledge bases.

Created by Shivam Saini as collage Assignment 🎓

Key Features

AI-Powered Workspace

- **Chat with PDF:** Upload any document and have a natural conversation with it. Ask questions, get summaries, and clarify doubts instantly.
- **Context-Aware:** The AI understands the full context of your uploaded materials.

Smart Study Tools

- **Flashcard Generator:** Automatically generate high-quality flashcards from your notes to reinforce memory. Supports iterative generation for precision.
- **Mock Test Generator:** Create adaptive quizzes with mixed question types (MCQ, True/False, Short Answer) to test your knowledge.
- **AI Note Taker:** Generate structured study notes from your documents in seconds.

Advanced Capabilities

- **Multi-Model Support:** Switch between **Google Gemini**, **OpenAI GPT-4o**, and **Ollama** (Local LLMs) seamlessly.
- **Real-time Streaming:** Experience instant feedback with character-by-character streaming responses.
- **Data Persistence:** All your chats, notes, flashcards, and test results are securely saved to the database.
- **Newsletter Subscription:** Stay updated with the built-in newsletter subscription feature.

Futuristic UI/UX

- **Glassmorphism Design:** A stunning, modern interface with glass effects and neon accents.
- **Responsive:** Fully optimized for all devices.
- **Smooth Animations:** Powered by Framer Motion for a fluid user experience.

Tech Stack

- **Framework:** [Next.js 15](#) (App Router)
- **Language:** TypeScript
- **Styling:** Tailwind CSS, Vanilla CSS (Variables)
- **Database:** MongoDB Atlas (Mongoose)
- **AI Integration:**
 - Google Generative AI SDK (Gemini)
 - OpenAI Node SDK
 - Ollama (Local LLMs)
- **UI Libraries:** Framer Motion, Lucide React, Sonner (Toasts)
- **PDF Handling:** pdf-parse, react-pdf

Key Libraries Used

Category	Library	Purpose
Core	<code>next , react</code>	Framework and UI library
AI	<code>@google/generative-ai</code>	Google Gemini API integration
AI	<code>openai</code>	OpenAI API integration
Database	<code>mongoose</code>	MongoDB object modeling
Animation	<code>framer-motion</code>	Complex animations and transitions
Icons	<code>lucide-react</code>	Beautiful, consistent icons
State	<code>zustand</code>	Lightweight state management
PDF	<code>react-pdf , pdf2json</code>	Rendering and parsing PDF files
Markdown	<code>react-markdown</code>	Rendering AI responses
Styling	<code>tailwind-merge , clsx</code>	Utility for dynamic classes
UI	<code>sonner</code>	Toast notifications
Scroll	<code>lenis</code>	Smooth scrolling experience
Upload	<code>react-dropzone</code>	Drag and drop file uploads



Getting Started

Prerequisites

- Node.js 18+ installed.
- A MongoDB Atlas account.
- API Keys for Google Gemini and/or OpenAI.
- (Optional) Ollama installed locally for local model support.

Installation

1. Clone the repository:

```
git clone https://github.com/yourusername/nebula-mind.git
cd nebula-mind
```

2. Install dependencies:

```
npm install
```

3. Set up environment variables:

Create a `.env.local` file in the root directory and add the following:

```
MONGODB_URI=your_mongodb_connection_string
GEMINI_API_KEY=your_gemini_api_key
OPENAI_API_KEY=your_openai_api_key
```

4. Run the development server:

```
npm run dev
```

5. Open the app:

Visit <http://localhost:3000> in your browser.



Contributing

Contributions are welcome! Please feel free to submit a Pull Request.



License

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